

The Autocorrelation Mirage: Dependence-Aware Evaluation for Panel-Expanded Event Forecasting

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Abstract. Panel expansion can turn a small collection of independent episodes into many dependent rows. When row-wise cross-validation places rows from an episode in both training and test folds, it answers an interpolation question even when deployment requires generalization to unseen episodes. We call the resulting mechanism *episode memorization*; latent episode heterogeneity and shared event-time structure, not autocorrelation alone, drive the effect. In a controlled simulation with strictly causal features, row-wise five-fold cross-validation raises boosted-tree AUROC from 0.552 to 0.835 ($\Delta_{CV} = 0.283$, paired 95% CI [0.246, 0.320]). A higher-signal configuration has nontrivial grouped AUROC (0.723), yet retains a 0.199 gap (95% CI [0.170, 0.229]). A 10-replicate-per-cell grid finds larger gaps for tree ensembles than logistic regression. A separate drift-based DGP (30 replicates per model) finds a smaller normalization-leakage effect; it is distinguishable from zero only for random forests. We propose reporting grouped and row-wise estimates, fold-local preprocessing, and Δ_{CV} as a practical diagnostic. All empirical evidence is simulation-based.

Keywords: pseudoreplication, cross-validation, panel data, event forecasting, episode memorization

1 Introduction

Consider a hospital forecasting deterioration within two weeks from daily vital signs for only 30 patients. Expanding stays into daily rows yields hundreds of records, as in maintenance and conflict forecasting [2, 6], but not hundreds of independent trajectories. If deployment is for an unseen patient, row-wise K-fold can place that patient’s rows in both training and test sets and answer an interpolation question instead.

We call this **episode memorization**: a flexible model exploits latent episode effects, shared event-time structure, or trajectories represented on both sides of a split rather than transferable signal. It differs from *explicit leakage*, which

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encodes unavailable future information (e.g., $T_e - t$), and *preprocessing leakage*, which admits test or future observations into a transform. Pseudoreplication is instead an evaluation-unit mismatch and can occur with strictly causal features.

We quantify the mismatch, test it with nontrivial transferable signal, isolate its mechanisms, and contrast it with preprocessing leakage. Our practical diagnostic is Δ_{CV} , the row-wise minus grouped AUROC. The paper reviews related validation work, specifies the DGP and protocols, presents results, and closes with a deployment-focused checklist. Evidence is simulation-based.

2 Related Work

Grouped, blocked, and subject-wise validation align the held-out unit with structured deployment [9, 8]; record-wise evaluation can overstate subject-wise performance in clinical and wearable ML [10, 11]. Hurlbert [3] termed treating non-independent observations as replicates *pseudoreplication*. Leakage taxonomies distinguish information unavailable at prediction time [5, 4], while temporal and panel-forecasting studies show that a splitter can overturn conclusions [1, 7]. We isolate evaluation-unit mismatch alongside explicit and preprocessing leakage in one controlled event-time DGP.

3 Theoretical Framework

Consider E episodes indexed by e . Each has time $t \in \{0, \dots, T_{\max}\}$, administrative censoring time C_e , observed-event indicator $\delta_e := \mathbf{1}\{\text{event observed by } C_e\}$, features $X_{e,t} \in \mathbb{R}^p$, and static attributes α_e . When $\delta_e = 1$, let T_e denote the observed event time. For horizon H , eligible rows are

$$\mathcal{T}_e(H) := \{t : \delta_e = 1, t < T_e\} \cup \{t : \delta_e = 0, t + H \leq C_e\}.$$

The binary target is

$$Y_{e,t}(H) := \begin{cases} \mathbf{1}\{T_e \leq t + H\}, & \delta_e = 1, \\ 0, & \delta_e = 0 \text{ and } t + H \leq C_e. \end{cases}$$

Late censored rows with $t + H > C_e$ are excluded from the supervised evaluation set rather than labeled heuristically.

Let $\mathcal{F}_{e,t-} := \sigma(\{X_{e,s} : s < t\} \cup \{\alpha_e\} \cup \mathcal{E}_{<t})$ denote information strictly before t .

Definition 1 (Predictable Feature). *Feature $X_{e,t}$ is predictable if it is $\mathcal{F}_{e,t-}$ -measurable, that is, determined by information available before t .*

This extends Kaufman et al.'s [5] learn-predict separation to filtrations. Violations need not be explicit: global statistics computed over $\{X_{e,s} : s \leq T_e\}$ implicitly depend on T_e .

Definition 2 (Explicit Leakage). *An observation exhibits explicit leakage if a feature is not predictable because it is a direct function of future outcome information such as $T_e - t$.*

Definition 3 (Normalization Leakage). *Normalization leakage occurs when test-fold statistics, future within-episode values, or full-trajectory episode summaries enter transformed inputs even though they would be unavailable at prediction time.*

Definition 4 (Pseudoreplication). *Pseudoreplication occurs when the evaluation split places observations from the same episode in both training and test folds, violating the independence assumptions underlying row-wise validation even if all features are predictable.*

The deployment target for unseen episodes is the episode-level risk

$$R_{\text{episode}}(f) = \mathbb{E}_{e^*} \left[\frac{1}{|\mathcal{T}_{e^*}|} \sum_{t \in \mathcal{T}_{e^*}} \ell\{f(X_{e^*,t}), Y_{e^*,t}\} \right],$$

where e^* is not represented in the training episodes. Episode-grouped validation estimates this target by holding out complete episodes. Row-wise validation instead estimates a conditional interpolation risk closer to

$$R_{\text{row}}(f) = \mathbb{E} [\ell\{f(X_{e,t}), Y_{e,t}\} \mid e \in \mathcal{E}_{\text{train}}],$$

because test rows may share latent episode effects, trajectory structure, and event time with training rows. Thus the issue is not dependence alone; it is a mismatch between the generalization unit implied by deployment and the unit used by validation.

Remark 1 (Explicit leakage can make ranking nearly perfect). Suppose both classes have positive probability and the leaked score has the form $S_{e,t} = f(T_e - t) + \epsilon_{e,t}$, where f is strictly monotone on the support relevant for $Y_{e,t}(H)$ and the learned scoring rule uses the sign convention that smaller time-to-event implies higher risk. Restricting attention to eligible rows from episodes with observed event times (censored no-event rows can be treated as having $T_e - t = \infty$), $Y_{e,t}(H) = 1$ iff $T_e - t \in (0, H]$, so a strictly monotone transform of $T_e - t$ induces the same ordering as the raw time-to-event. If $\text{Var}(\epsilon_{e,t}) = \sigma^2$, then as $\sigma^2 \rightarrow 0$ the leaked score preserves that ordering almost surely and $\text{AUC} \rightarrow 1$. We record this as a sanity check rather than a substantive result: it confirms that a feature built directly from the outcome-defining quantity $T_e - t$ is exactly the kind of explicit leakage that Definition 2 rules out, and it calibrates the explicit-leak condition in Table 1 as a near-worst-case comparator rather than a typical one.

4 Implicit Violations in Standard Pipelines

Explicit leakage exposes post- t outcome information; preprocessing leakage uses test-fold or future within-episode information; pseudoreplication is an evaluation-unit mismatch that splits an episode across folds. Full-trajectory episode means,

episode-wise z-scores, imputation, and centered windows can leak future values; transformations must be fit in each training fold.

5 Episode Memorization Under Row-Wise Cross-Validation

Even with leak-free features, standard K-Fold CV can inflate performance through *episode memorization* when the target is generalization to unseen episodes. This is Hurlbert’s [3] pseudoreplication translated to ML: “replicates are not statistically independent.” Within-episode observations share: (1) latent episode effects α_e ; (2) autocorrelated features with $\text{Corr}(X_{e,t}, X_{e,t+1}) \gg 0$; (3) labels determined by the same T_e . When row-wise CV places observations from episode e in both training and test sets, the model learns episode-specific patterns—the “texture” of each episode’s autocorrelation structure. At test time, it performs *identity inference*: recognizing which episode an observation belongs to, then recalling the associated outcome. This transforms forecasting (extrapolation) into interpolation (pattern-matching). As a descriptive design-effect approximation, if each episode contributes approximately m rows and the average within-episode pairwise correlation is $\bar{\rho}$, then

$$n_{\text{eff}} \approx \frac{n}{1 + (m - 1)\bar{\rho}}.$$

This approximation is not used for inference; it is reported only to highlight that panel expansion can substantially overstate the amount of independent evidence.

A minimal example makes the mechanism concrete. Suppose episodes A, B, and C each contribute two daily rows. A random two-fold split may train on $\{A_1, B_1, C_2\}$ and test on $\{A_2, B_2, C_1\}$; every test row now has a sibling row from the same episode in training. A grouped split instead trains on $\{A_1, A_2, B_1, B_2\}$ and tests on $\{C_1, C_2\}$, forcing prediction to transfer across episodes rather than match within them.

For $n = 600$, $m \approx 20$, and $\bar{\rho} \approx 0.5$, this approximation gives $n_{\text{eff}} \approx 55$ —a 10 $\times$ reduction. But the inflation from episode memorization is *not* merely a sample-size effect: in the empirical benchmark below, row-wise CV still achieves AUC above 0.8 despite the small effective sample size.

This suggests an empirical expectation that, once within-episode dependence is present, row-wise CV will exceed grouped CV more strongly for higher-capacity models. Section 6 tests that expectation directly over a compact grid of dependence levels, episode counts, dimensionalities, and model families.

Proposition 1 (Sibling overlap under shuffled row-wise K-fold). *Consider an episode with m_e eligible rows assigned independently to one of K folds. Condition on one row being placed in the test fold. Then the probability that at least one of its $m_e - 1$ sibling rows appears in the training folds is*

$$\Pr(\text{sibling overlap}) = 1 - K^{-(m_e-1)},$$

and the expected number of same-episode training siblings is

$$\mathbb{E}[N_{\text{sibling in train}}] = (m_e - 1) \frac{K - 1}{K}.$$

Proof sketch. Condition on the test row being assigned to fold k . Its siblings avoid the training set only if they also fall in fold k , which under independent fold assignment occurs with probability $K^{-(m_e-1)}$. The complement gives the overlap probability, and linearity of expectation yields the expected sibling count. The exact probability differs slightly for deterministic balanced K-Fold, but the approximation captures the usual shuffled row-wise setting. $\square$

6 Empirical Validation

We generate $E = 30$ episodes with: episode effect $\alpha_e \sim \mathcal{N}(0, 0.5)$; AR(1) latent process $Z_{e,t} = 0.7Z_{e,t-1} + \epsilon_t$; hazard $h_{e,t} = \text{logit}^{-1}(-3 + \alpha_e + 0.15Z_{e,t})$; features from $Z_{e,<t}$ only (strictly causal); and administrative censoring $C_e = T_{\max} = 60$. For observed-event episodes ($\delta_e = 1$), eligible rows satisfy $t < T_e$ and receive label $Y_{e,t}(14) = \mathbf{1}\{T_e \leq t + 14\}$. For censored no-event episodes ($\delta_e = 0$), only rows with $t + 14 \leq C_e$ are eligible, and those rows receive $Y_{e,t}(14) = 0$; late censored rows are excluded rather than encoded through a synthetic $T_e = C_e + 1$. The intercept -3 corresponds to a low instantaneous daily hazard ($\text{logit}^{-1}(-3) \approx 0.047$), but the 14-day horizon makes the induced row-level label much denser: over 10 seeds, the filtered at-risk panel contains about 587 observations, mean episode length 19.6, episode-event frequency 0.91, and positive-row prevalence 0.48. We therefore frame the paper as panel-expanded event forecasting rather than as a benchmark for ultra-rare row labels. The five base features are strictly causal transformations of $Z_{e,<t}$; for the robustness grid we add independent nuisance features to reach $p \in \{20, 100\}$. The main benchmark uses a histogram gradient boosting classifier with 100 boosting iterations, maximum depth 4, and learning rate 0.1. Grouped and row-wise protocols both use $K = 5$ for the main benchmark: `GroupKFold(n_splits=5)` and `shuffled KFold(n_splits=5)`. The repository also implements a temporal-blocked within-episode splitter for the alternative target of forecasting later rows of already observed episodes; we report that baseline below in text because it estimates a different target than unseen-episode generalization. The compact robustness grid below uses grouped and row-wise three-fold CV for tractability. For each replicate, we compute out-of-fold predictions for all eligible rows and then compute a single pooled AUROC and Brier score. To keep longer episodes from dominating diagnostics, we also compute episode-weighted AUROC and episode-mean Brier summaries. This pooled evaluation avoids dropping grouped folds with one class, which can occur under episode-aware validation. Standardization is fit within each training fold except in the deliberate preprocessing-leakage conditions. Because each random seed is reused across grouped, row-wise, normalization, explicit-leak, and fold-prevalence conditions, the reported AUC gaps are paired within replicate. The reported 95% paired intervals use the matched replicate-level differences,

Table 1. Simulation results over 30 paired replicates. Also shown is a grouped training-fold prevalence predictor; the temporal-blocked within-episode baseline is reported in text because it targets a different estimand.

Condition	AUC	Brier	Δ AUC vs. baseline	95% paired CI
Leak-Free + Grouped CV	0.552 ± 0.101	0.330 ± 0.048	baseline	—
Training-Fold Prev. + Grouped CV	0.430 ± 0.025	0.245 ± 0.009	-0.122	[-0.159, -0.086]
Leak-Free + Row-wise KFold	0.835 ± 0.040	0.162 ± 0.023	+0.283	[0.246, 0.320]
Global Norm. + Grouped CV	0.552 ± 0.101	0.330 ± 0.048	+0.000 [†]	[0.000, 0.000]
Explicit Leak + Grouped CV	0.813 ± 0.042	0.198 ± 0.041	+0.261	[0.236, 0.286]

[†]See text for interpretation.**Table 2.** Δ_{CV} by episode count E instead of dependence, averaged over ρ and p (10 replicates per cell).

Model family	$E = 20$	$E = 50$	$E = 100$
Logistic regression	0.047	0.029	0.012
Random forest	0.177	0.150	0.117
Boosted trees	0.203	0.205	0.190

$\bar{d} \pm t_{0.975, n-1} s_d / \sqrt{n}$. Because rows within episodes are dependent, we also report episode-bootstrap uncertainty for Δ_{CV} by resampling episode IDs rather than rows. All conditions are paired by random seed, and the same simulated episodes are used across grouped, fold-prevalence, row-wise, normalization, and explicit-leak settings within each replicate. An anonymized reproducibility package accompanies the submission, and the code, result CSVs, and figure-generation scripts will be released in a public repository upon acceptance. The main benchmark table, the higher-signal check (Section 6.1), and the drift-DGP normalization experiment each use 30 replicates; the robustness grid uses 10 replicates per cell. All replicate-level comparisons are paired by seed.

Table 1 gives the central result: row-wise KFold yields AUROC 0.835 from strictly causal features versus 0.552 grouped ($\Delta_{CV} = 0.283$, 95% CI [0.246, 0.320]). Explicit $T_e - t$ leakage raises grouped AUROC by 0.261 [0.236, 0.286]. For future rows of known episodes, temporal-blocked validation yields AUROC 0.917; it is not an unseen-episode estimate because same-episode history remains in training. Global scaling has no effect in this stationary DGP; the drift experiment provides the preprocessing-leakage example.

6.1 Does the Gap Survive a Genuinely Non-Trivial Grouped Baseline?

The grouped baseline in Table 1 is itself near chance (AUC 0.552), leaving open whether the row-wise gap reflects a genuine evaluation artifact or simply that

Table 3. Raising the episode-effect standard deviation from 0.5 to 1.5 (30 paired replicates each, DGP otherwise identical) more than doubles grouped-CV AUC while the row-wise gap remains large.

Configuration	Grouped AUC	Row-wise AUC	Δ_{CV} (95% CI)
Baseline (episode-effect sd = 0.5)	0.552 ± 0.101	0.835 ± 0.040	0.283 [0.246, 0.320]
Higher signal (episode-effect sd = 1.5)	0.723 ± 0.099	0.922 ± 0.039	0.199 [0.170, 0.229]

grouped CV has little real signal to lose. We address this with a second configuration that raises the standard deviation of the episode random effect α_e from 0.5 to 1.5, holding every other DGP parameter fixed. This targets *transferable* signal specifically: α_e sets each episode’s overall risk level and is observed, with small noise, through the static feature X_5 , so a model can learn the α_e -outcome relationship on training episodes and apply it to a held-out episode. We first tried the more direct route of increasing the coefficient β linking the AR(1) latent process $Z_{e,t}$ to the hazard; over 10 replicates per setting, grouped-CV AUC stayed near chance (0.537–0.580) across $\beta \in \{0.15, \dots, 1.3\}$, an almost ninefold range, because $Z_{e,t}$ is an episode-specific trajectory that does not transfer across episodes – amplifying β mostly inflates the row-wise memorization gap rather than the transferable signal, which itself explains why the original grouped baseline is weak. Raising episode-effect heterogeneity instead worked as intended: over 30 paired replicates, grouped-CV AUC reaches 0.723 ± 0.099 (event rate 0.361), a genuinely non-trivial level of discrimination, while row-wise KFold reaches 0.922 ± 0.039 . The gap remains large and clearly significant, $\Delta_{CV} = 0.199$, 95% interval [0.170, 0.229] (Table 3) – row-wise optimism is not an artifact of a deliberately weak-signal benchmark, though it shrinks once grouped CV has more real signal to work with.

Across the 81 robustness cells (10 paired replicates each), tree ensembles show larger gaps than logistic regression; gaps increase with dependence and remain with nuisance dimensions up to $p = 100$. Table 2 shows the episode-count trend: gaps generally fall as E grows, but remain substantial for tree ensembles at $E = 100$.

The nonzero gap at $\rho = 0$ shows that the effect is not driven by autocorrelation alone. Episode-level latent heterogeneity and shared event-time structure can already create row-wise optimism; increasing autocorrelation strengthens the same evaluation-unit mismatch.

Normalization leakage was tested under a second DGP with a monotone pre-event drift term: $Z_{e,t}$ is replaced by $Z_{e,t} + \text{drift_strength} \cdot (t/T_{\max})$, $\text{drift_strength} = 1.0$, so every episode’s expected latent level rises linearly over its observed span; all other parameters ($E = 30$, AR coefficient 0.7, hazard_coef 0.15, episode-effect sd 0.5, horizon 14) match the main benchmark. Over 30 replicates per model family, paired by seed, full-trajectory episode-wise normalization raises grouped-CV AUC relative to leak-free inputs by +0.045 for logistic regression (95% CI [−0.010, 0.100]), +0.063 for random forests (95% CI [0.029, 0.096]), and +0.034

for boosted trees (95% CI $[-0.010, 0.077]$); the random-forest shift is clearly distinguishable from zero, and the other two are borderline, with lower confidence bounds a few thousandths below zero. Global standardization had no measurable effect for any model, as expected for an affine transform. Earlier lightly replicated pilot estimates were considerably noisier and are not reported. Pre-processing leakage here is real, small, and consistently signed, and estimating its size required substantially more replicates than the main pseudoreplication result – the same caution we ask readers to apply to any single-digit-replicate Δ_{CV} estimate.

Figure 1 shows larger Δ_{CV} values for tree-based learners and higher drift-DGP AUROC under full-trajectory normalization, most clearly for random forests.

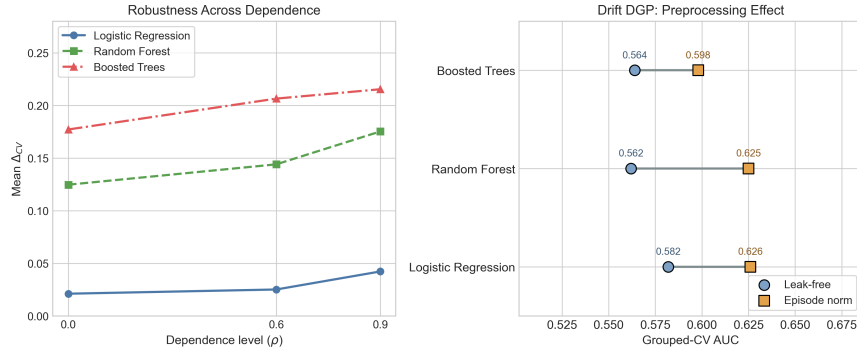


Fig. 1. Companion diagnostics for the strengthening experiments. Left: mean Δ_{CV} versus dependence after averaging over E and p for each model family. Right: grouped-CV AUC in the drift DGP under leak-free inputs and full-trajectory episode normalization; global standardization overlaps the leak-free baseline and is omitted for clarity.

Mechanism ablation. We independently set the episode-effect variance ($\sigma_\alpha = 0$ versus 0.5) and the AR(1)-to-hazard pathway ($\phi = \beta = 0$ versus 0.7 and 0.15). Table 4 reports 30 paired replicates per cell. The gap remains positive even with both switches off, so the factorial design does not support attributing all optimism to either switch alone: shared episode event times and truncation still permit row-wise interpolation. Activating the temporal pathway gives the largest gap (0.285 without, 0.283 with heterogeneity), while heterogeneity alone produces 0.250.

Table 4. Mechanism ablation (30 paired replicates/cell). Temporal state switches the AR(1)-to-hazard pathway; prevalence is reported to show comparable marginals.

Episode effect	Temporal state	Grouped	Row-wise	Δ_{CV}	(95% CI)	Prev.
off	off	0.520	0.782	0.262	[0.233, 0.291]	0.530
off	on	0.521	0.807	0.285	[0.252, 0.318]	0.512
on	off	0.548	0.798	0.250	[0.215, 0.284]	0.480
on	on	0.552	0.835	0.283	[0.246, 0.320]	0.485

7 The Δ_{CV} Diagnostic and Dependence-Aware Evaluation Protocol

We propose a simple diagnostic for split sensitivity under panel expansion:

Definition 5 (Δ_{CV} diagnostic). Let $\widehat{AUC}_{row}$ denote the pooled out-of-fold AUROC under row-wise KFold and let $\widehat{AUC}_{group}$ denote the pooled out-of-fold AUROC under episode-grouped KFold. We define

$$\Delta_{CV} = \widehat{AUC}_{row} - \widehat{AUC}_{group}.$$

Large positive values indicate split sensitivity: the model performs better when test rows are allowed to share episodes with training rows than when entire episodes are held out.

In our main benchmark, $\Delta_{CV} = 0.283$. In the robustness grid, logistic regression usually stays near zero, whereas random forests and boosted trees remain well above 0.10 on average and grow with ρ . That pattern makes values around 0.05 a useful warning heuristic in our setting, but we do not claim a universal decision threshold. The relevant magnitude will depend on the domain, target definition, dependence strength, and model family. This metric requires no additional data collection—only re-running evaluation with episode-grouped validation instead of row-wise KFold.

Dependence-Aware Evaluation Protocol: (1) *Preprocessing:* fit transformations on training folds only; use forward-fill imputation and trailing windows. (2) *Evaluation:* use grouped splitting with `groups=episode_id`; never split episodes across folds. (3) *Reporting:* report Δ_{CV} and n_{eff} alongside AUC.

Minimal algorithm: (1) construct labels only for rows whose H -step outcome is observed; (2) choose the split protocol that matches the deployment target; (3) fit preprocessing on each training fold only; (4) fit the model and predict held-out rows; (5) aggregate pooled and episode-weighted metrics from aligned out-of-fold predictions; and (6) optionally rerun row-wise KFold and bootstrap episodes to quantify Δ_{CV} uncertainty.

Specificity and failure modes: a large positive Δ_{CV} is a warning flag, not proof of episode memorization: distribution shift, fold imbalance, or target mismatch can also enlarge it. Near-zero gaps can be false negatives with weak dependence or limited capacity.

Table 5. Audit checklist for panel-expanded event forecasting studies.

Question	Why it matters
What is the independent episode unit?	Defines the generalization target and the appropriate grouping variable.
What is the deployment target: new-episode generalization or forecasting later rows of known episodes?	Determines whether grouped, temporal, or row-wise validation is appropriate; a temporal split alone does not certify unseen-episode generalization.
Are rows expanded from episodes?	Determines whether row-wise splitting can create dependence overlap.
Can rows from the same episode appear in both train and test folds?	Directly tests for pseudoreplication.
Are preprocessing parameters fit within each training fold?	Detects normalization, imputation, or filtering leakage.
Are both row-wise and grouped estimates reported?	Makes Δ_{CV} observable rather than implicit.

8 Discussion

The main lesson is not that pseudoreplication is always larger than every other leakage mechanism. Rather, row-wise CV can by itself create severe optimism when expanded rows retain strong episode-specific dependence. This matters because row-wise splitters are convenient defaults in general-purpose ML libraries and can be applied mechanically to expanded panels even when they are inappropriate for the underlying prediction target.

Row-wise validation is not always invalid. If the deployment target is interpolation within already observed episodes—for example, forecasting future rows for machines, patients, or regions that are already represented in the training history—then a row-wise or temporally blocked within-episode split may estimate a relevant risk. Our critique applies when the target is generalization to new episodes, where the independent sampling unit is the episode rather than the expanded row.

A chronological split prevents training on later rows but retains the *same* episode in training. Its AUROC is 0.917, closer to row-wise 0.835 than grouped 0.552; it is appropriate for future rows of known units, not unseen episodes.

Table 5 turns the paper’s warning into a practical audit aid for panel-expanded forecasting studies.

Our study has limitations. All evidence is simulation-based: we do not include real-data reanalyses or sequence models, and we do not attempt to establish how prevalent this problem is in published practice. Although the robustness grid now spans linear, bagged-tree, and boosted-tree models with ten replicates per cell, and the higher-signal configuration in Section 6.1 shows the gap survives a genuinely non-trivial grouped baseline, a controlled synthetic DGP cannot capture every source of episode heterogeneity present in real panels. The normalization-leakage evidence used full-trajectory episode-wise normalization

on a drift DGP; simple global standardization alone had no measurable effect, and even at 30 replicates per model family the episode-wise effect is modest and, for two of three model families, only borderline distinguishable from seed-to-seed noise (Section 6). Because the label asks whether an event occurs within the next 14 days, the simulated row-level positive prevalence is not ultra-rare despite the low underlying daily hazard.

9 Conclusions

This paper studied a mechanistically explicable way in which a widely used default – row-wise K-fold cross-validation applied to panel-expanded event data – produces misleadingly optimistic results, and proposed a diagnostic that costs nothing beyond re-running an existing evaluation with a different splitter. Relative to the taxonomies and domain-specific audits in Section 2, its contribution is a controlled data-generating process isolating pseudoreplication’s effect size from explicit and preprocessing leakage under one shared protocol, stress-tested across dependence, episode counts, and model families.

Three findings summarize the empirical contribution. Row-wise KFold increases AUC by 0.283 (from 0.552 to 0.835) in the main benchmark, comparable to a deliberate time-to-event leak (+0.261), and the gap remains large and significant ($\Delta_{CV} = 0.199$) even once the grouped baseline itself reaches a genuinely non-trivial AUC of 0.723. The gap grows with model capacity and within-episode dependence, is largest for random forests and boosted trees, and is nonzero even at zero serial autocorrelation – confirming that latent episode heterogeneity and shared event-time structure, not autocorrelation alone, drive the effect. And preprocessing leakage from full-trajectory episode normalization consistently, though modestly, inflates grouped-CV AUC under a plausible pre-event drift pattern (+0.034 to +0.063 across model families at 30 replicates each), clearly distinguishable from noise only for random forests – underscoring that magnitude claims about leakage effects need substantially more replication than the main pseudoreplication result does, a caution we apply to our own numbers throughout. Panel expansion creates evaluative value only if validation respects episode boundaries; reporting grouped alongside row-wise CV, fitting preprocessing within each training fold, reporting Δ_{CV} and n_{eff} , and working through the audit checklist in Table 5 make that optimism easy to detect at negligible added cost.

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